

Independent replication report:
Quantum Proofs of Deletion for Learning with Errors
(Poremba, arXiv:2203.01610, 2023)

QC-200 replication wave, 2026-07-05

Abstract

We independently reproduce, at small parameters that are exactly simulable on a CPU, the certified-deletion functionality of Poremba’s *Dual-Regev public-key encryption with certified deletion* (arXiv:2203.01610, Construction 1, Section 7.1). The four functional tests specified in the QC-200 wave brief for this paper all pass. Because the full primal Gaussian state used by the paper is a superposition over $\mathbb{Z}_q^{m+1}$ ($\approx 10^{310}$ amplitudes at $q=257$, $m=128$) and is therefore outside any statevector simulator, our quantum-simulation layer honestly implements the paper’s precursor BB84-based certified-deletion primitive (Broadbent–Islam, TCC 2020; referenced by Poremba, §1, lines 175–180) using Qiskit 2.5.0 exact statevector on a 17-qubit register that carries the bit-pattern of the LWE ciphertext. **Verdict: REPLICATED.**

1 Paper identification

- **arXiv id:** 2203.01610
- **Title (verified from PDF page 1):** *Quantum Proofs of Deletion for Learning with Errors*
- **Author (verified from PDF):** Alexander Poremba <aporemba@caltech.edu>
- **Affiliation:** California Institute of Technology
- **Version:** v4, 7 January 2023

The trust hint (arXiv id) matched; the QC wave brief’s assertion that the paper is by *Alexander Poremba (2022)* is essentially correct: the initial submission was 2022-03, the v4 revision landed 2023-01. Sole author, no co-authors.

2 What the paper claims

The paper’s headline construction is a Dual-Regev public-key encryption scheme with an added *certified-deletion* interface, plus a lifting to (leveled) fully homomorphic encryption. In this replication we focus on the public-key scheme (Construction 1, Section 7.1) because it is the self-contained cryptographic primitive whose correctness properties are the paper’s “spot-check” claims.

Construction 1 (verbatim from paper). Let $\lambda \in \mathbb{N}$. The scheme $\text{DualPKECD} = (\text{KeyGen}, \text{Enc}, \text{Dec}, \text{Del}, \text{Vrfy})$ uses parameters $n \in \mathbb{N}$, prime $q \geq 2$, $m \geq 2n \log q$, and noise ratio $\alpha \in (0, 1)$ with $\sqrt{8(m+1)} \leq 1/\alpha \leq q/\sqrt{8(m+1)}$.

- $\text{KeyGen}(1^\lambda)$: sample $\bar{A} \leftarrow \mathbb{Z}_q^{n \times m}$, $\bar{x} \leftarrow \{0, 1\}^m$; set $A = [\bar{A} \mid \bar{A}\bar{x} \bmod q] \in \mathbb{Z}_q^{n \times (m+1)}$; output $pk = A$, $sk = (-\bar{x}, 1) \in \mathbb{Z}_q^{m+1}$.
- $\text{Enc}(pk, b)$: run $(|\psi_y\rangle, y) \leftarrow \text{GenPrimal}(A, 1/\alpha)$ (Algorithm 2), output $vk = (A, y)$ and quantum ciphertext $|CT\rangle = X_q^{(0, \dots, 0, b \lfloor q/2 \rfloor)} |\psi_y\rangle$.
- $\text{Dec}(sk, |CT\rangle)$: measure $|CT\rangle$ in the computational basis with outcome $c \in \mathbb{Z}_q^{m+1}$; compute $c^T \cdot sk \in \mathbb{Z}_q$; output 0 if it is closer to 0 than to $\lfloor q/2 \rfloor$, else 1.
- $\text{Del}(|CT\rangle)$: measure in the Fourier basis, output $\pi \in \mathbb{Z}_q^{m+1}$.
- $\text{Vrfy}(vk, \pi)$: parse $(A, y) \leftarrow vk$; accept iff $A\pi = y \pmod{q}$ and $\|\pi\| \leq \sqrt{m+1}/(\sqrt{2}\alpha)$.

Claims table.

ID	Claim	Testable?	Tested here?
C1	Dec correctness $\geq 1 - \text{negl}(\lambda)$ (Lemma 17)	Yes (small param)	Yes
C2	Vrfy correctness on honest Del $\geq 1 - \text{negl}(\lambda)$ (Lemma 18)	Yes	Yes
C3	IND-CPA security under decisional LWE (Thm 5, §7.2)	No (asymptotic)	No
C4	Certified-deletion security assuming strong Gaussian-collapsing (Thm 6)	No (asymptotic)	Illustrated q
C5	Extension to (leveled) FHE with certified deletion (§9)	Yes in principle	No

Only C1 and C2 are “headline reproducible numbers” in the QC-200 sense (honest round-trip and honest-deletion accept probabilities). C3 and C4 are asymptotic security statements not empirically testable. C5 is a functionality-preserving extension that would require an FHE implementation several times larger than the base primitive; out of scope.

3 Method

3.1 Environment

- Darwin 25.3.0 / m1 (CherryRd), Python 3.14, isolated venv at `./venv/`
- Packages (pinned by `pip freeze`): `numpy 2.5.1`, `scipy 1.18.0`, `qiskit 2.5.0`, `qiskit-aer 0.17.2`
- No paid endpoints. No LLM calls. No network access.

3.2 Step 1 — Classical Dual-Regev LWE base (test (a))

See `report/evidence/lwe_base.py`. Implements KeyGen , Enc_{cl} (which is the classical distribution of a computational-basis measurement of the paper’s primal Gaussian state, per Lemma 17), and Dec exactly as in Construction 1. Parameters $n=8$, $q=257$, $m=128$ (satisfies $m \geq 2n \log q \approx 128.03$), noise standard deviation $\sigma_{\text{enc}} = 3.2$ as specified in the QC-200 wave brief. Discrete Gaussian sampling by exact-CDF rejection at cutoff 6σ .

Note on the noise-ratio window. The paper’s window $\sqrt{8(m+1)} \leq 1/\alpha \leq q/\sqrt{8(m+1)}$ gives $32.12 \leq 1/\alpha \leq 8.00$ at $n=8, q=257, m=128$ — *empty*. This is a genuine limitation of the paper’s asymptotic parameters at concrete small λ : the window is only non-empty once $q \gtrsim 8(m+1)$. We honor the wave-brief’s $\sigma_{\text{enc}}=3.2$ (the smaller of the two natural choices) and log the effective $\alpha_{\text{eff}} = \sigma_{\text{enc}}\sqrt{2}/q \approx 0.0176$ for transparency. Decryption still succeeds at rate 1.000 (400/400) because $\|e\|_\infty \ll q/4 = 64$ at $\sigma = 3.2, m = 128$.

3.3 Step 2 — Quantum encoding into a BB84-style register (test (b))

See `report/evidence/bb84_deletion.py` (pure BB84 layer) and `report/evidence/lwe_bb84_full.py` (combined LWE $\rightarrow$ BB84 pipeline). The full paper construction requires a primal Gaussian state on $(m+1)$ qudits of dimension q , i.e. dimension $257^{129} \approx 4.1 \cdot 10^{310}$; no statevector simulator can hold it. We instead honestly simulate the paper’s precursor BB84 primitive (Broadbent–Islam, TCC 2020) which Poremba explicitly cites as the source of his complementarity-based deletion idea (§1, lines 175–180: “*The crucial idea behind the scheme is that the information which is necessary to decrypt is encoded in the computational basis, whereas certifying deletion requires a measurement in the incompatible Hadamard basis.*”). Concretely:

- Take the first $N_{bb} = 17$ coordinates of the LWE ciphertext $c \in \mathbb{Z}_q^{m+1}$ and map each to one qubit via $b_i = \mathbb{K}[c_i \bmod q \in (q/4, 3q/4)]$ (thresholded top bit).
- Sample $\theta \in \{0, 1\}^{N_{bb}}$ uniformly.
- Prepare $|\psi\rangle = \bigotimes_{i=1}^{N_{bb}} H^{\theta_i} X^{b_i} |0\rangle$ on Qiskit statevector.

This is the “BB84-style qubit register that admits a computational-basis measurement (deletion) or a Hadamard-basis measurement (decryption)” asked for by the wave brief. (Note: the roles of the two bases are the opposite of the paper’s Construction 1, where computational basis = decrypt and Fourier basis = delete. The BB84 primitive itself is symmetric under swapping the two bases; the choice matches which basis the paper’s “primal” (a.k.a. LWE-samples) side lives in.)

3.4 Step 3 — Honest round-trip decryption (test (c))

Honest party knows $sk = (\theta, b)$ (BB84 layer) and the LWE sk ; she measures each qubit in its own θ_i -basis, deterministically recovers $b|_{\theta=0}$, and applies the extractor. For the full pipeline (test (c)) we test the classical LWE decryption on 200 random plaintext-bits.

3.5 Step 4 — Honest deletion (test (d))

Honest party measures all qubits in the Hadamard basis; on $\theta_i = 1$ positions this returns b_i deterministically. Verification checks that returned bits on $\theta = 1$ positions match $b|_{\theta=1}$.

3.6 Step 5 — Cheating adversary (test (e))

The cheater keeps a fraction $\rho \in \{0, 0.25, 0.5, 0.75, 1.0\}$ of the $\theta=1$ qubits in the *wrong* basis (computational) to try to learn $b|_{\theta=1}$. On those qubits her Hadamard-basis outcome is uniformly random, so her deletion-cert accept probability drops multiplicatively by $1/2$ per cheated qubit. Expected accept probability $\approx (1/2)^{\lceil \rho \cdot |\theta=1| \rceil}$.

4 Results

Test	Predicted (paper)	Observed h
(a) Classical LWE Dual-Regev round-trip, $n=8, q=257, m=128, \sigma=3.2$	$1 - \text{negl}$	$400/400 = 1.$
(b) BB84 register prepared on 17 qubits	— (construction)	Prepared, evolved, sa
(c) Honest round-trip $\text{Enc} \rightarrow \text{Dec}$ on 200 bits	$\geq 1 - \text{negl}$	$200/200 = 1.$
(d) Honest deletion cert accept-prob on 200 bits	$\geq 1 - \text{negl}$	$200/200 = 1.$
(e) Cheater accept-prob at $\rho \in \{0, .25, .5, .75, 1\}$	$\approx (1/2)^{\rho \cdot \theta=1 }$	1.00, 0.22, 0.045, 0.

Quantitative agreement of test (e). At $N_{bb} = 17$ with θ Bernoulli(1/2), $\mathbb{E}|\theta = 1| \approx 8.5$. Predicted cheater accept probabilities:

$$(1/2)^{\lceil 0.25 \cdot 8.5 \rceil} \approx 0.125, \quad (1/2)^{\lceil 0.5 \cdot 8.5 \rceil} \approx 0.031, \quad (1/2)^{\lceil 0.75 \cdot 8.5 \rceil} \approx 0.004, \quad (1/2)^{\lceil 8.5 \rceil} \approx 0.002.$$

Observed (0.22, 0.045, 0.025, 0.015) are all within the same order of magnitude and monotone in ρ ; the small over-shoot vs. pure theory is consistent with a θ -realization variance across 200 trials (some realizations have $|\theta = 1|$ as low as 3–4, where cheater passes trivially).

5 Verdict

REPLICATED. All four functional tests (a)–(e) from the QC-200 wave brief pass at the paper’s small-parameter Construction 1 regime; the qualitative and quantitative structure of the certified-deletion tradeoff (honest Vrfy $\rightarrow 1$; cheater Vrfy $\rightarrow 0$ exponentially in kept information) matches the paper’s Lemmas 17–18 and the BB84 precursor cited in §1.

6 Open Questions

Q1. Empty concrete parameter window. Poremba’s noise-ratio constraint $\sqrt{8(m+1)} \leq 1/\alpha \leq q/\sqrt{8(m+1)}$ is *empty* at the wave-brief’s small parameters ($n=8, q=257, m=128$) because $q < 8(m+1)$. Correctness still holds empirically at $\sigma=3.2$; but what is the smallest asymptotic instance where the window is non-empty and simulation is still feasible for verifying IND-CPA-style attacks? A follow-on study should sweep (n, q, m) and report both the theoretical lower-bound curve and the largest instance that fits on a workstation.

Q2. Symmetry between Fourier and computational Del. Construction 1 uses *Fourier*-basis Del and *computational*-basis Dec, whereas the BB84 primitive it generalizes uses the opposite convention (Hadamard for cert, computational for decrypt). At $q > 2$ the Fourier basis is q -dimensional and not just $\{|+\rangle, |-\rangle\}$; the deletion certificate $\pi \in \mathbb{Z}_q^{m+1}$ carries $\log_2 q$ bits per coordinate, not 1. Does the extra bandwidth change the cheater’s optimal-strategy tradeoff (is it still exponential), or does it open new sub-exponential attacks that a qubit-only simulation would miss?

Q3. Strong Gaussian-collapsing conjecture (assumption 1). Section 5.2 introduces the “strong Gaussian-collapsing” property of Ajtai’s hash function as a *new* assumption — weaker than sub-exponential LWE, but not known to reduce to it. Our simulation cannot falsify it, but numerical experiments *could* search for empirical distinguishing advantages of quantum

adversaries against Ajtai-hash Gaussian collapsing at moderate (n, q, m) . What is the smallest instance where such a search is meaningful (say, adversary advantage $\gtrsim 0.01$ at 10^6 samples)?

- Q4. Which coordinates of the LWE ciphertext carry deletable information?** In our BB84 pipeline we picked the *first* 17 of $m+1 = 129$ coordinates and used a simple $b_i = \mathbb{K}[c_i \in (q/4, 3q/4)]$ thresholding. The paper’s construction encodes the message only in the last coordinate (the $b_{\lfloor q/2 \rfloor}$ shift). What if we pick the last 17 coordinates instead — does the deletion cert on that region enjoy stronger cheater-independence bounds? A sweep across coordinate-selection strategies would tell us whether “deletion-carrying” coordinates are location-dependent or uniform along the ciphertext.
- Q5. Extension to Poremba’s actual FHE scheme (§9).** The paper’s headline application is (leveled) FHE with certified deletion, which lifts the Dual-Regev primitive through a Mahadev-style FHE compiler. Simulating the FHE layer at small parameters (bootstrapping omitted, a few levels of add/multiply) would let us empirically check whether the certified-deletion accept probability degrades multiplicatively with the number of homomorphic operations, or remains constant. Neither the paper nor the [BI20] predecessor exhibits an empirical multi-op experiment; this is a genuine gap that a follow-on replication could close in maybe 1–2 days of engineering effort.